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Enhanced microbial diversity in the saliva microbiome induced by short-term probiotic intake revealed by 16S rRNA sequencing on the IonTorrent PGM platform

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ABSTRACT

Microbial communities populating several human body habitats are important determinants of human health. Cultivation-free community-wide approaches like bacterial 16S rRNA sequencing recently revolutionized the study of such human-associated microbial diversity, and the continuously decreasing cost/throughput ratio of current sequencing platforms is further enhancing the availability and effectiveness of microbiome research. The IonTorrent PGM platform is among the latest available commercial high-throughput sequencing tools, but it is just starting to be used for 16S rRNA surveys with only episodic assessments of its performance. We present here the first IonTorrent profiling of the human saliva microbiome collected from 12 healthy individuals. In this cohort, a subset of the volunteers was asked to assume a probiotic product, in order to investigate its impact on the composition and the structure of the saliva microbiome. Analysis of the generated dataset suggests the suitability of the IonTorrent platform for 16S rRNA surveys, even though some platform-specific choices are required to optimize the consistency of the obtained bacterial profiles. Interestingly, we found a marked and statistically significant increase of the overall bacterial diversity in the saliva of individuals who received the probiotic product compared to the control group, suggesting a short-term effect of probiotic product administration on oral microbiome composition.

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1. Introduction

Human-associated microbial communities comprise a wide range of microorganisms from the three kingdoms of life (Bacteria, Archaea and Eukarya) as well as from viruses, and are now widely recognized as important determinants of human health. In particular, the microbiome of the oral cavity is the second most diverse microbiome of the human body, after the colon (The Human Microbiome Project Consortium, 2012), and has been quite comprehensively investigated in healthy subjects (Dewhirst et al., 2010; Segata et al., 2012; Wade, 2013). It shows high spatial heterogeneity, reflecting various environmental conditions and physico-chemical differences between structures of the oral cavity (Segata et al., 2012). The oral microbiome plays an important role in maintaining oral and systemic health, primarily by inhibiting colonization by pathogens through the phenomenon of colonization resistance (Santagati et al., 2012; Vollaard and Clasener, 1994). Community shifts have been also associated with diseases (Burne et al., 2012; Curtis et al., 2011; Junemann et al., 2012; Liu et al., 2012; Wade, 2013; Yang et al., 2012), with the peculiarity that the agents of the most common oral diseases (dental caries and periodontitis) are members of the commensal microbiome.

The oral microbiome does not seem to be largely influenced by geographical differences (Nasidze et al., 2009), suggesting that the diet and the environment play only a minor role in its composition. Indeed, with the exception of fermentable carbohydrates obtained from the host's diet, the primary source of nutrients required by the oral communities are endogenous glycoproteins from saliva

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and gingival crevicular fluid (Rudney et al., 2010). In this context, probiotic interventions with specific bacterial strains (usually from *Bifidobacterium*, *Lactobacillus* and *Streptococcus* genera) have been proposed as modulators for the saliva microbiome (Dierksen et al., 2007; Horz et al., 2007; Snel et al., 2011; Wescombe et al., 2009; Yli-Knuuttila et al., 2006), but these studies typically focus on tracking the persistence of the probiotic strain after intake and the correlation with clinical data, without investigating the overall changes in the saliva microbiome structure induced by the probiotic product. Commercial probiotic products have been tested as modulators for the whole gut microbiome with reported therapeutic properties in mice (Martin et al., 2008; Veiga et al., 2010), but only one microbiome-wide assessment of the effect of oral probiotic has been performed so far (Burton et al., 2013).

Assessing in vivo the composition of microbiomes has been made possible by cultivation-free community-wide sequencing approaches which have recently revolutionized the study of such human-associated microbial diversity. In particular, the advent of high-throughput sequencing has made bacterial 16S rRNA surveys accessible and particularly appealing for its cost-effectiveness trade-off (Hamady and Knight, 2009). Since its recent availability, microbiome research has been mostly technology-driven and especially cost-throughput sensitive as the investigators aim at analyzing larger cohorts with increased sensitivity and depth. One of the most recent high-throughput sequencing platforms, the IonTorrent PGM (Rothberg et al., 2011), is just starting to be used for this kind of studies, and promises to be particularly suited to 16S rRNA profiling. IonTorrent performances for 16S rRNA sequencing are being thoroughly assessed, with a few studies available on human-associated microbiomes, including two for the gut (Milani et al., 2013; Petrof et al., 2013), one for the dental plaque (Junemann et al., 2012), the hand skin (Rosenthal et al., 2014) and the airways (Salipante et al., 2013). Some environmental microbiome studies on this platform are also available (Whiteley et al., 2012; Yergeau et al., 2012).

Here we present the first IonTorrent PGM profiling of the human saliva microbiome and we assess the impact on the overall saliva microbiome structure of a short-term probiotic intervention. To these ends, we sequenced the bacterial 16S rRNA of fourteen saliva samples from healthy individuals, subjecting a subset of these to intake of a commercial probiotic product. We describe the composition of the microbial communities in these samples and the issues related to the use of the IonTorrent PGM technology; furthermore, we discuss the impact of short-term probiotic intake on samples taxonomic composition and metabolic structure.

2. Materials and methods

2.1. Study design

Saliva samples were collected from 12 adult volunteers (6 males and 6 females) and processed for investigating their microbial composition. All 12 volunteers were Italian healthy individuals who did not receive any systemic or oral antibiotic treatment in the last 12 months prior to the beginning of the study. Volunteers were asked not to clean their teeth neither with toothpaste, nor toothbrush nor antiseptic mouthwash on the day of sample collection (i.e. at least 14 h before sample collection). A subset of 5 subjects out of 12 underwent a probiotic-free diet on the day of sample collection. This group included one subject who provided 2 saliva samples employed as biological replicates. The remaining subset of 7 subjects received 100 g of a commercial probiotic product containing milk fermented with *Streptococcus thermophilus*, *Lactobacillus delbrueckii* subsp. *bulgaricus*, and *Lactobacillus paracasei* strains and supplemented with vitamin B6 and vitamin D. Four subjects in this group received the probiotic at breakfast (at least 6 h before sample collection) while the other 3 subjects received the same product at lunch (at least 1 h before sample collection) (see Table 1). The study was approved by the Ethical Committee of the University of Trento and informed written consent was obtained from all volunteers.

2.2. Sample collection and DNA extraction

Each participant performed a mouth rinse with drinking water 30 min before sample collection. A total of 13 saliva samples were collected by passive drooling in a sterile 50 ml tube. Samples were vigorously mixed using the vortex and processed within 30 min from sample collection. Total DNA was extracted from 600 μl saliva using the QIAamp DNA Blood Mini Kit (QIAgen s.r.l., Milano, Italy), following the body fluid spin protocol, provided by the manufacturer, with the following modifications. Cell lysis treatment time was extended to 20 min for all samples. DNA bound to the QIAamp membrane was washed 3 times, the last one with 70% molecular-grade ethanol and DNA was eluted in 100 μl molecular-grade water and immediately processed. DNA yields and purity were determined measuring absorbance at 260 nm, A260/A280 and A260/A230 ratios with the Nanodrop 2000 spectrophotometer (Nanodrop products, Wilmigton, USA).

2.3. 16S rRNA amplicon library preparation

The V3 hypervariable region of the 16S rRNA gene was amplified by polymerase chain reaction (PCR) with the Platinum PCR SuperMix High Fidelity (Invitrogen; Life Technologies Europe BV, Monza, Italy), employing 20 ng template DNA and each primer at a final concentration of 200 nM in 50 μl final reaction volume. The following thermal cycling conditions were used on a T1 96-well plate thermocycler (Biometra GmbH, Goettingen, Germany): 95 °C 5 min initial denaturation; 95 °C 30 s, 55 °C 30 s, 72 °C 90 s for 28 cycles; 72 °C 10 min final elongation.

13 different sets of HPLC-purified primers were employed, one set per individual. Each set included a barcoded forward primer, comprising at its 5′ the IonTorrent adaptor A and at its 3′ a sequence recognizing the V3 16S rRNA region (CCTACGGGRSGCAGCAG, (Milani et al., 2013)). A unique reverse primer sequence was used for each set, containing at its 5′ the IonTorrent adaptor trP1 and at its 3′ a sequence targeting the V3 region of the 16S rRNA (ATTACCGCGGCTGCT, (Milani et al., 2013). The length of the amplified region corresponds to 162 nucleotides. We used the following samples-specific barcodes (from sample 1 to sample 13): TCTGGATGAC, TCTAGAGGTC, TAGGTGGTTC, CTGACCGAAC, TGAGCGGAAC, TTCCGATAAC, TTCGTGATTC, CTGCAAGTTC, CAGAAGGAAC, TACCAAGATC, AAGAGGATTC, TAAGGAGAAC, CTAAGGTAAC.

PCR products were analyzed by electrophoresis on a 1.5% agarose gel in TAE 1x and amplicon library products were extracted from gel and purified with the Wizard SV Gen PCR Clean-Up System (Promega Italia s.r.l., Milano, Italy), as described by the manufacturer. The amplicon libraries were further purified, in order to remove residual primer dimers and any contaminant, using the Agencourt AMPure XP DNA purification kit (Beckman Coulter s.r.l., Milano, Italy) following manufacturer's instructions. Samples were eluted in 50 μl molecular-grade water. Amplicon libraries' concentration, size range and purity were measured using the Nanodrop 2000 spectrophotometer, the dsDNA high sensitivity assay kit on the Qubit 2.0 fluorometer instrument (Invitrogen; Life Technologies Europe BV, Monza, Italy), and the Agilent high sensitivity (HS) DNA Kit on the Bioanalyzer 2100 instrument (Agilent Technologies Italia S.p.A., Milano, Italy).

Table 1
Sequencing and preprocessing statistics for the 16S rRNA samples. The * symbol indicates biological replicates. Probiotic administration column indicates whether the sample had probiotic before collection; the Probiotic administration time column indicates when the probiotic was administered (either Breakfast or Lunch of collection day).

Sample name	Probiotic administration	Probiotic administration time	Total reads	High quality reads	Reduced by (%)
1*	No	–	91,261	29,302	67.89
2*	No	–	104,048	34,120	67.21
3	No	–	209,578	53,581	74.43
4	No	–	216,388	28,511	86.82
5	Yes	Lunch	214,061	63,919	70.14
6	Yes	Lunch	186,699	45,136	75.82
7	No	–	283,975	56,590	80.07
8	Yes	Breakfast	337,633	52,875	84.34
9	Yes	Lunch	248,150	52,547	78.82
10	Yes	Breakfast	191,858	55,109	71.28
11	No	–	197,498	52,854	73.24
12	Yes	Breakfast	209,418	57,492	72.55
13	Yes	Breakfast	193,205	43,088	77.70

2.4. 16S rRNA amplicon sequencing

From the concentration estimated using Qubit (Invitrogen; Life Technologies Europe BV, Monza, Italy) HS kit (Part #Q32854) and the average size of each amplicon library, we calculated the amount of DNA fragments per microliter. Each library was subsequently diluted to 9 nM and pooled. Prior to clonal amplification the library pool was diluted to 26 pM according to the Fusion-method protocol instructions.

Sample emulsion PCR, emulsion breaking, and enrichment were performed using the Ion OneTouchTM 200 Template Kit v2 DL (Life Technologies Europe BV, Monza, Italy; Part #4480285 Rev. 6.0), following manufacturer's instructions. Briefly, the input concentration of the DNA template copy/Ion Sphere Particles (ISPs) was added to the emulsion PCR master mix and the emulsion generated using a One Touch DL (Life Technologies Europe BV, Monza, Italy). Next, ISPs were recovered and template-positive ISPs enriched using Dynabeads MyOne Streptavidin C1 beads (Life Technologies Europe BV, Monza, Italy). ISP enrichment was confirmed using the Qubit 2.0 fluorometer (Invitrogen; Life Technologies Europe BV, Monza, Italy), and the sample was prepared for sequencing using the Ion Sequencing 200 Kit protocol (Life Technologies Europe BV, Monza, Italy; Part # 4474004 Rev.G). The complete sample was loaded on an Ion 316 chip and sequenced on the PGM for 500 cycles.

2.5. Data analysis

The FASTQ file generated by the IonTorrent PGM platform was converted to FASTA and QUAL files with Mothur (Schloss et al., 2009). Initial sequence analysis was performed with QIIME (Caporaso et al., 2010): after demultiplexing and primer removal, reads were selected by average Phred quality score greater than 25 and for the absence of homopolymers longer than 6 bases. Subsequently, only reads with a minimum length of 160 nucleotides were retained, and trimmed when longer than 170 nucleotides. Operational Taxonomic Units (OTUs) were built at 94% and 97% sequence identity with uclust (Edgar, 2010) as further discussed in the Results; only OTUs with at least ten assigned reads were retained, and the most abundant sequence for each OTU was selected as representative. Taxonomic assignment to the lowest possible rank was performed with RDP (Wang et al., 2007), using the Greengenes (DeSantis et al., 2006) reference dataset (gg_13_5 release). Alpha diversity (observed_species, PD_whole_Tree and chao1) and beta diversity measures (Lozupone and Knight, 2005; Whittaker, 1972) (weighted and unweighted UniFrac metrics displayed through 2D PCoA) were computed by QIIME with rarefaction to prevent statistical artefacts due to different sequencing depths. Biomarker discovery in probiotic takers versus non-takers samples was performed with LEfSe (Segata et al., 2011) with default parameters and LDA log score threshold set to 2, and sample-specific prediction of the metabolic potential of the microbial communities was performed with PICRUSt (Langille et al., 2013) followed by processing with HUMAnN (Abubucker et al., 2012) (for metabolic pathway summaries) and QIIME (for visualization). Circular cladogram visualization was performed using GraPhlAn (https://bitbucket.org/nsegata/graphlan/).

2.6. Data availability

Raw sequences were deposited in the NCBI Short Read Archive (http://www.ncbi.nlm.nih.gov/sra) and are available with the following accession numbers: SAMN02402850, SAMN02402851, SAMN02402852, SAMN02402853, SAMN02402854, SAMN02402855, SAMN02402856, SAMN02402857, SAMN02402858, SAMN02402859, SAMN02402860, SAMN02402861, SAMN02402862 and SAMN02402863.

3. Results and discussion

3.1. IonTorrent sequencing of the saliva microbiome

The 13 saliva samples collected from 12 healthy subjects were subject to DNA extraction and amplification of the variable region V3 of the 16S rRNA gene prior to sequencing on an IonTorrent PGM platform (316 chip). The sequencing run produced ~4.8 million reads that were reduced to ~3 million usable reads after removing wells containing polyclonal reads. The obtained sequences were then demultiplexed, quality filtered, and analyzed with QIIME (Caporaso et al., 2010). The number of retained high-quality reads was considerably in excess of 20 thousand (median 52.7 thousand, maximum 63,919 reads for sample 5) for all. Percentage of removed reads for the samples ranged from 67% to 87%, with 75% as median level of reduction (see Table 1). These percentages are higher than the typical read removal rates for other sequencing technologies (e.g. 5–25% typical removal rate for 454 pyrosequencing (Kunin et al., 2010)) and should be taken into consideration when designing microbiome surveys with IonTorrent. A minor fraction of the reads (~4%) resulted to be longer than the expected amplicon size; upon inspection the first part of the reads (up the amplicon size) contained valid 16S rRNA sequences, whereas the additional nucleotides did not show strong homology with any 16S regions and were considered chimeric sequencing products. Read trimming to 170 nucleotides was thus required in order to avoid the introduction of additional noise in the OTU picking and taxonomic assignment steps, and also for this aspect future 16S rRNA sequencing surveys on the IonTorrent platform should explicitly consider this technology-dependent artefact removal step.

3.2. Taxonomic composition and complexity of the saliva microbiome as seen by IonTorrent PGM

We then taxonomically profiled the saliva microbial communities (see Section 2) summarizing the relative abundances of microbial clades at different taxonomic levels (Fig. 1A). We identified five phyla making up >1% of the community in at least one sample with two phyla, Firmicutes and Proteobacteria, dominating the communities with varying relative abundance (24.3–90.2% and 4.3–73.5% respectively). Bacteroidetes were also consistently present in all samples, but to a lesser extent (1.1–14.2%). Other lower-abundance phyla include Actinobacteria and Tenericutes as well as members of the Spirochaetes phylum which is thought to include several oral pathogenic organisms, e.g. *Treponema* strains (Dahle et al., 1993), and so-called "environmental" phyla such as SR1 were also detected. At the genus level (Fig. 1B), *Streptococcus* and *Neisseria* make up for almost 60% of the total reads, with a peak of almost 80% in sample 11. *Veillonella*, *Prevotella*, *Selenomonas* and *Haemophilus* are also found in all samples with uneven quantitative distributions. Bacilli, Lactobacillales, and Streptococcaceae are the most abundant class-, order-, and family-level clades respectively, as reported in Supplementary Fig. S1. Abundances for all these taxonomic levels in our samples are reported in Supplementary Tables 1A, 1B, 1C, 1D and 1E. The two biological replicates (samples 1 and 2) are found to display a remarkably similar taxonomic composition both at the phyla and at the genus level, suggesting a very high reproducibility of microbiome profiles with the IonTorrent platform.

We compared our profiles with the large 16S rRNA survey (166 samples) performed by the Human Microbiome Project (HMP) (Segata et al., 2012; The Human Microbiome Project Consortium, 2012) on the same body habitat. Fig. 2 displays an overall consistency between the two datasets at the phylum level, despite the geographical distance of the two cohorts (U.S. versus Italy) and the difference in the sequencing technology (Roche 454 vs. IonTorrent PGM). Firmicutes, is the most abundant phylum in both cohorts with a similar level of consistency at the genus level, with *Streptococcus*, *Haemophilus*, *Veillonella*, *Prevotella* and *Neisseria* as the main taxonomic components. Furthermore, we noticed the presence of a considerable amount of *Lautropia* (0.1–4.3%) in our profiles, a genus which is not observed in the HMP cohort for the same body habitat, even though it was already known to be a commensal organism in the oral cavity (Dewhirst et al., 2010); *Lautropia* has been previously associated with gingivitis (*Lautropia* sp. HOTA94 (Kistler et al., 2013)) and found to antagonize pathogenic species in subgingival dental plaque (Fujinaka et al., 2013). Average genus-level relative abundances for our work, HMP saliva and HMP subgingival plaque are available in Supplementary Tables 2A, 2B and 2C. We also compared our results with the only other study that performed an IonTorrent 16S rRNA sequencing survey with samples from the oral cavity (Junemann et al., 2012); such survey was applied on the subgingival plaque microbiome which is known to be taxonomically distinct from the saliva microbiome (Segata et al., 2012): this indeed shows a change in the most abundant phyla, which are Bacteroidetes and Firmicutes. Firmicutes are thus shared as highly abundant in the two datasets. Comparison of Jünemann et al. dataset with HMP subgingival plaque confirms the presence of a difference in Bacteroidetes abundance, as can be observed for the saliva, which may be due both to cohort and platform differences.

To investigate the impact of a potentially higher noise level in IonTorrent PGM sequencing compared to more established sequencing technologies as noticed elsewhere (Loman et al., 2012), we repeated the computational pipeline by filtering OTUs at different minimum number of reads (2, 5, 10, 20, 50) and at two OTU picking identity thresholds (94% in addition to the conventional 97%, Supplementary Fig. 2). Alpha diversity (number of observed OTUs) (Whittaker, 1972) for this combination of parameters is reported for all samples in Fig. 3A: a trend towards a plateau in the number of observed species can be first noticed after 20 thousand reads possibly as a consequence of losing low-abundance OTUs when using high OTU cardinality cutoff. We eventually selected 10 as a minimum number of reads and 97% as identity threshold because these appear to be a good trade-off between overestimated noise-inflated alpha diversity (as when using 2 reads as OTU filtering threshold) and underestimated microbial richness due to low-abundance OTUs removal (as when using 50 reads as OTU filtering threshold). Although the 97% threshold is a rather arbitrary value when considering genus-specific 16S evolution rates and when accounting for the IonTorrent error rate, we still maintained this value as it is the de facto standard for 16S rRNA surveys (Hamady and Knight, 2009; Junemann et al., 2012; Milani et al., 2013; Petrof et al., 2013; Rosenthal et al., 2014) but we increased the minimum required number of sequences for each valid OTU (10) to ensure consistency of diversity analysis. Alpha diversity curves

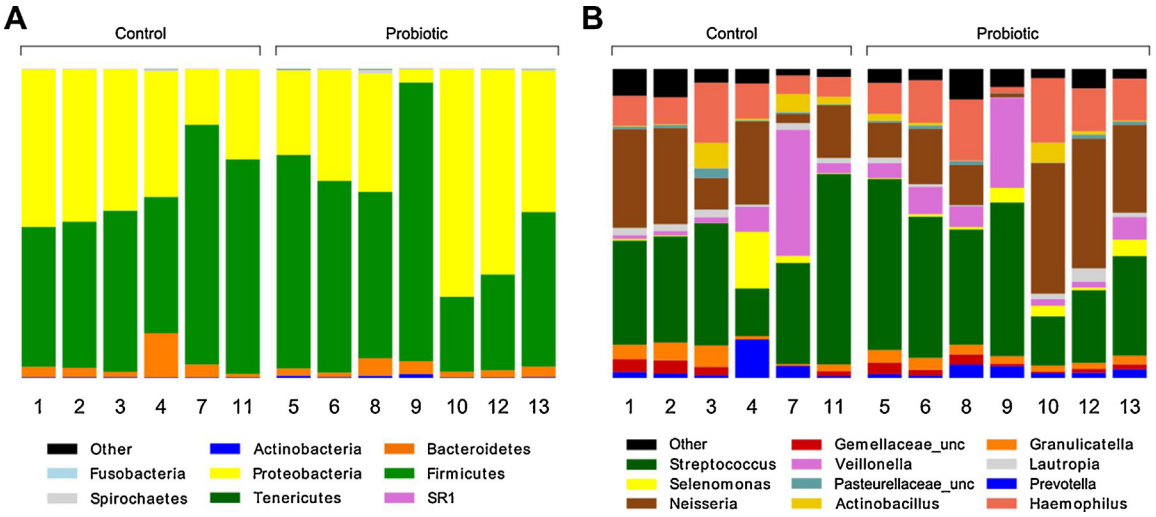


Fig. 1. Taxonomic composition of the 16S rRNA samples. (A) Relative abundances of the microbial phyla highlights the major role played by Firmicutes and Proteobacteria (green and yellow) in the microbiome, and to a less extent by Bacteroidetes (orange). (B) At the genus level, *Streptococcus*, *Haemophilus*, *Veillonella* and *Neisseria* (dark green, pink, yellow and brown) are shown to be the most abundant clades. Genera accounting for less than 0.5% of the samples are grouped in the "Other" category.

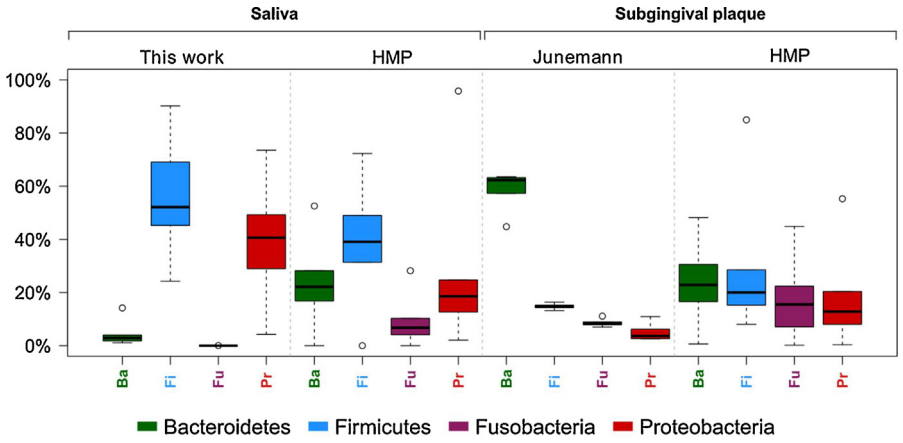


Fig. 2. Taxonomic composition comparison of Saliva and Subgingival Plaque datasets. The boxplot shown here compare the levels of the four most abundant phyla (Bacteroidetes, Firmicutes, Fusobacteria and Proteobacteria) in two saliva (this work and HMP Saliva profiling) and two subgingival plaque datasets (Jünemann et al., 2012 and HMP Subgingival Plaque). The two saliva and the two subgingival plaque datasets are shown to be consistent, with the subgingival plaque having Bacteroidetes instead of Firmicutes as most abundant phyla.

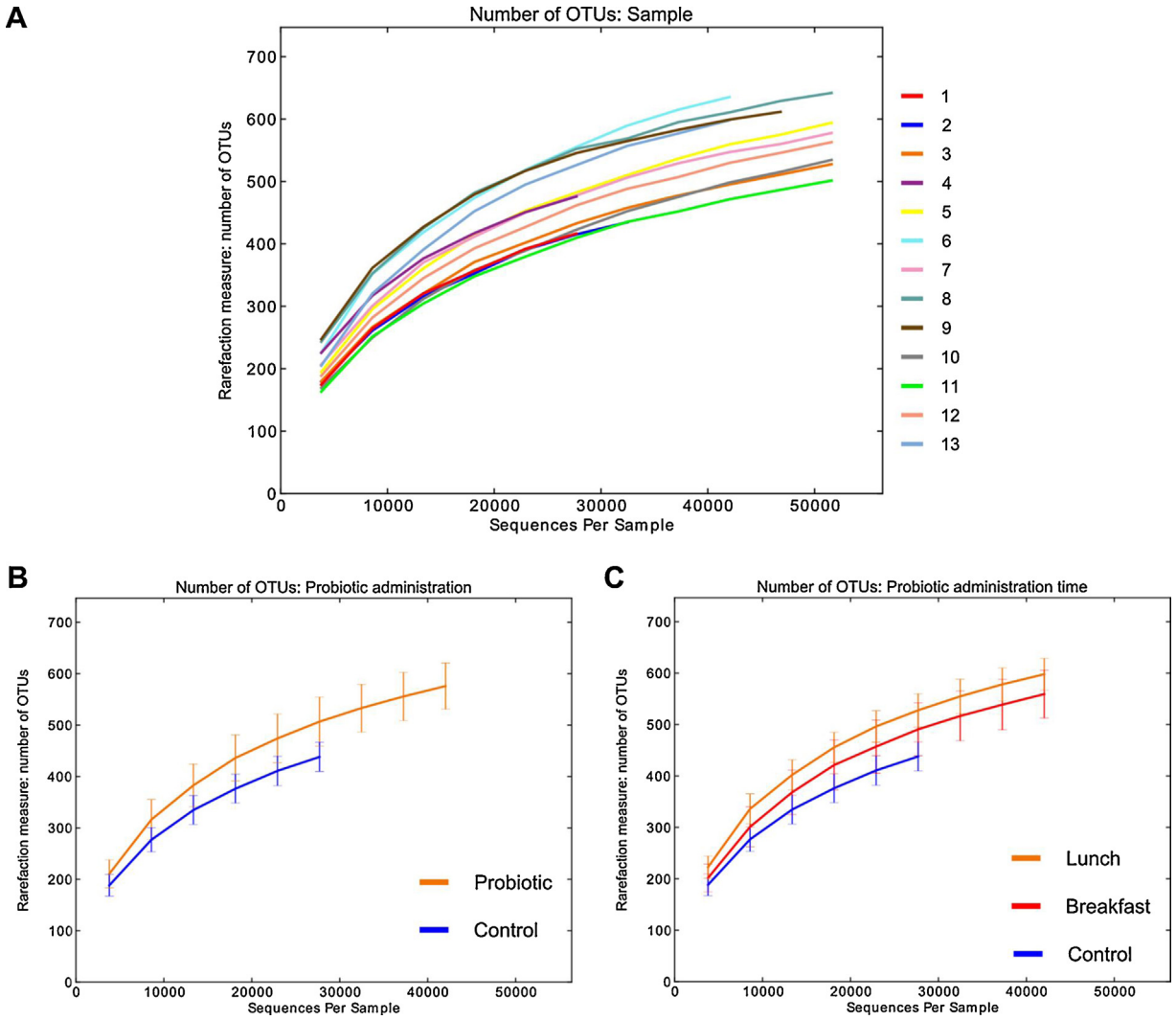


Fig. 3. Saliva microbiomes after probiotic intake are more diverse than control samples. Alpha diversity curves of single samples are summarized in condition-specific groups. Specifically, (A) shows alpha diversity (i.e. within sample complexity) plotted at increasing subsampled sequence sets for each sample in terms of number of observed OTUs (see Supplementary Fig. S3 for other alpha-diversity metrics). A trend towards a plateau can be first noticed after twenty thousand reads, suggesting this as the minimum number of reads needed for a sample to properly represents its microbial diversity by IonTorrent 16S sequencing; (B) reports the alpha diversity curves for probiotic versus control samples, highlighting the presence of a significant difference in complexity between these two groups (p-value 0.011), with probiotic samples characterized by more complex microbial communities than control ones. When the alpha diversity curves for probiotic intake is split by administration time; (C) the difference in complexity is retained with respect to control samples, with a less marked difference between breakfast and lunch administration time.

were almost identical for the two biological replicates confirming the reproducibility of the sequencing procedure with respect to the estimated community diversity.

3.3. Short-term probiotic intake significantly increase the complexity of the community

We next proceeded to study potential differences between samples which underwent probiotic intake (*S. thermophilus*, *L. delbrueckii* subsp. *bulgaricus*, and *L. paracasei* strains) and control samples. Interestingly, alpha diversity estimates on the two conditions are distinct with a statistically significant increase in complexity after probiotic intake (Fig. 3B, *p*-value 0.011 using the parametric *t*-test, *p*-value 0.014 using the non-parametric Wilcoxon rank-sum test). The microbial community diversity inversely correlates with the time of the probiotic intake (breakfast versus lunch) with richer saliva communities in subjects who received the probiotic product at lunch time (1 h before sample collection) although the difference is not statistically significant mainly due to the limited number of samples per condition (Fig. 3C). Different alpha-diversity measures (number of observed OTUs, Chao1 and Phylogenetic Diversity whole tree, see Supplementary Fig. S3) are all consistently detecting the probiotic group as having significantly more complex microbial communities.

To understand whether the phylogenomic and quantitative structure of the microbiome was also at play between the two samples groups, we performed a principal coordinate analysis (PCoA) on the taxonomic profiles. When using the non-quantitative unweighted UniFrac beta-diversity measure (Lozupone and Knight, 2005)) as underlying metric, the clustering analysis is suggestive for a separation between the two groups (Fig. 4A). However, if we consider the relative abundances of species in the beta diversity metric (using the weighted UniFrac measure) the comparison is strongly influenced by the dominating microbial clades and no probiotic intake-driven clusters are detectable. Furthermore, it should be observed that the explained variation in the unweighted UniFrac analysis is low (only 15.78% for the first component with respect to 55.79% in the weighted UniFrac analysis), suggesting a limited ability of this measure to describe the clustering patterns between our samples. Overall, this confirms the hypothesis of richer microbial communities associated with short-term probiotic intake even though the communities quantitative distribution of the main microbial components are likely not affected by standard probiotic products.

The two *Lactobacillus* strains present in the probiotic product (*L. delbrueckii* subsp. *bulgaricus*, and *L. paracasei*) were not detected in probiotic-intake samples. Indeed, we observe the presence of a very low-abundance OTU attributable to the *Lactobacillus* genus only in Sample 13. The presence of the probiotic strain *Streptococcus termophilus* in the samples could not be verified, as the 16S rRNA diversity within the *Streptococcus* genus does not permit species-level profiling and the OTUs of this genus are dominating all samples. However, to understand whether the intake of the *Streptococcus* strain was the main and direct reason for the observed increase in communities complexity, we repeated the alpha-diversity analysis after removing all *Streptococcus* OTUs from

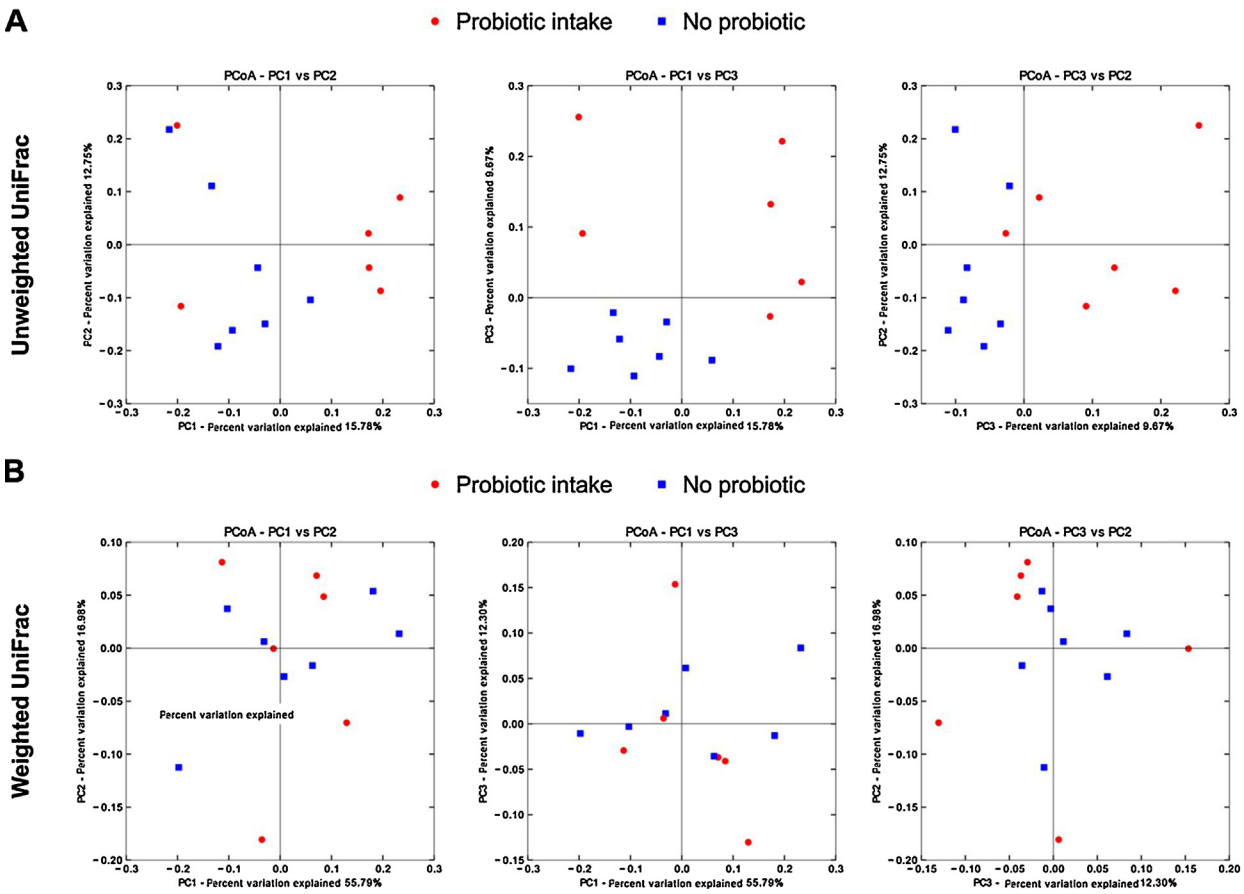


Fig. 4. The overall structure of the saliva microbiome is not affected by short-term probiotic intake. (A) A weak clustering is observed in PCoA space when the unweighted UniFrac measure is used as beta-diversity measure although the explained variance is low in this case. However, when the phylogenetic similarity is coupled with the quantitative abundance patterns using the weighted UniFrac metric (B), no apparent clustering trend can be noticed, suggesting that a change in relative abundances of the main components of the microbiome is not the driving feature characterizing probiotic and control samples.

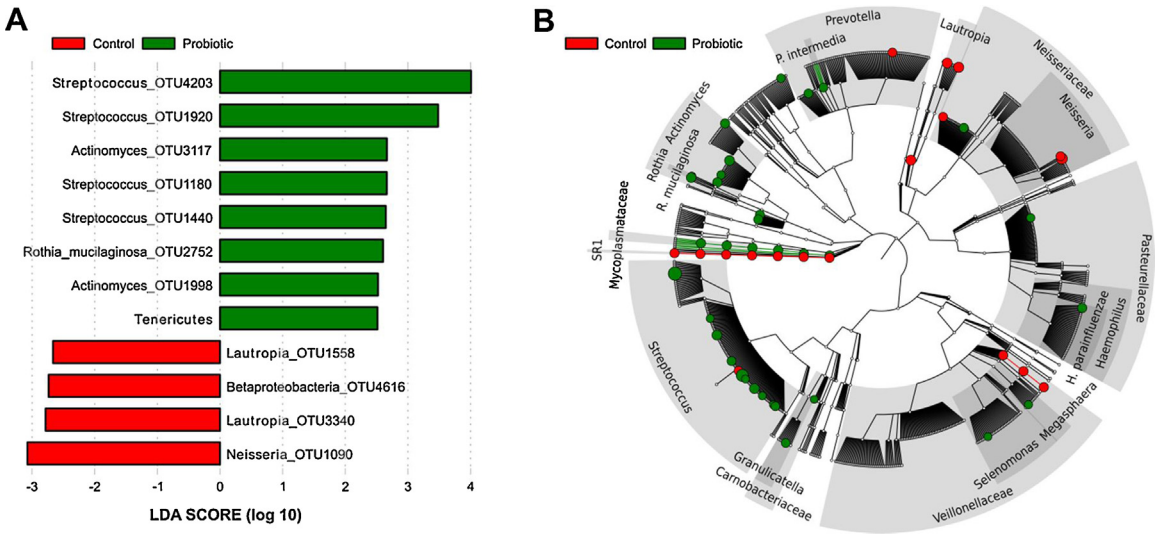


Fig. 5. Several *Streptococcus* and *Actinomyces* OTUs characterize the microbiome after short-term probiotic intake. Linear discriminant analysis with effect size highlights microbial clades able to differentiate probiotic and control samples. The OTUs showing a logarithmic effect size larger than 2.5 (A) are mainly members of the *Streptococcus* and *Actinomyces* genera (higher in probiotic-intake samples) and of the *Neisseria* and *Lautropia* genera (higher in control samples). When all differential OTUs (effect size threshold by default at 2.0) are displayed on the taxonomic tree of all the clades present at least once in the dataset (B), biomarkers for probiotic intake can be identified including several *Streptococcus* species as well as Actinomycetales organisms (*Rothia* and *Actinomyces*). Control samples are characterized instead by increased abundances of organisms from the environmental phylum SR1 and from the genus *Lautropia*.

the OTU table (Supplementary Fig. S4). The results indicate that the difference between the two sample groups still persists after the removal of this genus (p-value 0.035), thus suggesting that the probiotic strains alone cannot explain the increased diversity of the microbiome, and that some additional probiotic-induced mechanisms are responsible for the observed community-wide variations.

3.4. *Streptococcus, Actinomyces, and Rothia characterize the microbiome after short-term probiotic intake*

In order to identify which microbial clades, if any, could explain the increased diversity of short-term probiotic intake with respect to control samples, we performed a linear discriminant analysis effect size study using LEfSe (Segata et al., 2011), aimed at identifying robust biomarkers in the microbiome. Interestingly, as displayed by Fig. 5, several organisms robustly characterize the two conditions. Specifically, OTUs in the *Streptococcus* and the *Actinomyces* genera and in the *Rothia mucilaginosa* species have a significant LDA log score (at a threshold of 2.5) and characterize samples with short-term probiotic intake, while *Lautropia* and *Neisseria* OTUs are significant at the same effect size threshold for discriminating samples lacking probiotic administration. The distribution of these significant features on the bacterial taxonomic tree of all OTUs present in at least one of the samples (Fig. 5B), indicates that there is no single genus that can be considered responsible for the observed differences between probiotic and control samples. Interestingly, nine distinct *Streptococcus* OTUs are overabundant in the probiotic group, but a quantitatively enriched OTU in the same genus for the control group suggests a strain-level ecological modification induced by the probiotic product. In addition to significant variation for the mentioned known oral commensals supported by multiple distinct OTUs in the corresponding clades, several other OTUs show increased abundance in the probiotic group including those in the SR1 phylum, and in the *Selenomonas*, *Megasphera*, *Granulicatella* and *Haemophilus* genera (Fig. 5B). Taken together, this again suggests that while *Streptococcus* represents a significant component in probiotic samples change with respect to control, this genus cannot be the only responsible, with several other genera contributing to the induced observed effect.

3.5. *The metabolic structure of the saliva microbiome is not affected by short-term probiotic intake*

We eventually sought to understand whether short-term probiotic intake was impacting the overall metabolic potential of the microbial communities. We thus applied PICRUSt (Langille et al., 2013) in combination with HUMAnN (Abubucker et al., 2012) to infer the presence of metabolic pathways based on the set of annotated genomes selected for their consistency with the 16S rRNA profiles. The results of this analysis (Fig. 6) suggest that no significant change in the presence and abundance of metabolic functions is taking place between probiotic and control samples and no significant difference can be detected even within the two probiotic sub-groups. At the relatively high level of the functional categories accessible by these tools, transport-related pathways are the most variable across samples reflecting the differences in the metabolic specialization in the oral microbial species, whereas common core pathways related for example to the ribosomal machinery or the purine/pyrimidine metabolisms are very stable as expected. Overall, this analysis suggests that the functional structure of the saliva microbiome is stable and not affected by short-term probiotic intake. Consequently, we can hypothesize that the probiotic product is acting on low-abundance organisms that contributes only marginally to the overall metabolic potential of the community (as also the beta-diversity suggests) or that the microbial turnover induced by the probiotic product regards microbial clades occupying the same metabolic niche. Interestingly, an opposite phenomenon of relatively high taxonomic stability with low metabolic stability has been observed with respect to the intestinal microbiome associated with Inflammatory Bowel Diseases (Morgan et al., 2012). However, in healthy subjects, stability of metabolic functions with respect to carriage of microbial taxa has been confirmed by shotgun metagenomics for other oral cavity locations such as the supragingival plaque, the tongue dorsum, and the buccal mucosa (The Human Microbiome Project Consortium, 2012).

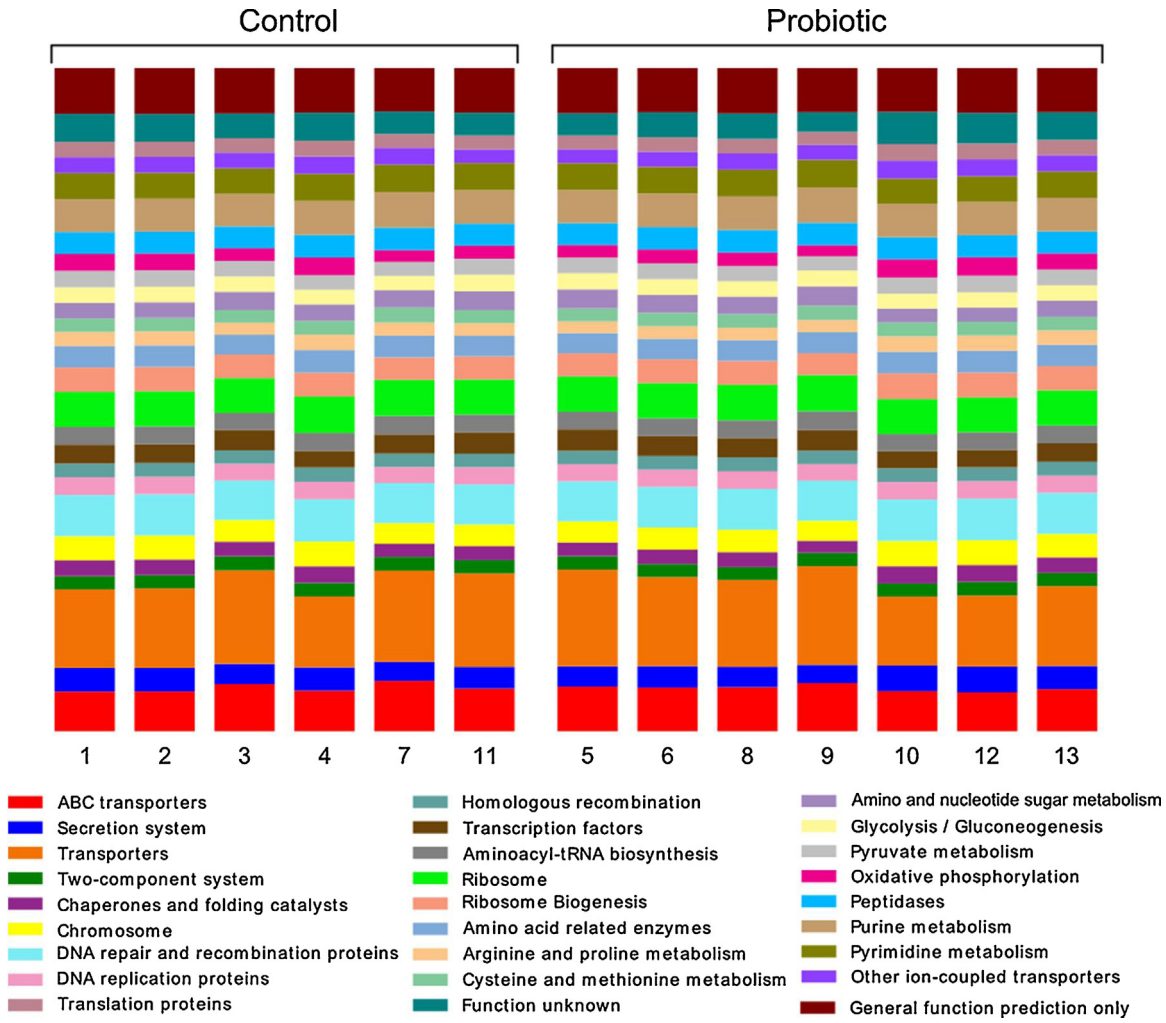


Fig. 6. The metabolic structure of the saliva microbiome is not affected by short-term probiotic intake. The most abundant KEGG pathways describing the overall metabolic potential of the microbiomes are reported in the samples, and they show a remarkably high consistency across samples. No significant differences between probiotic and control samples can be identified at the metabolic level.

4. Conclusions

We presented the first 16S rRNA high-throughput profiling of the human saliva microbiome with the IonTorrent PGM technology. Compared to the other sequencing platforms currently used for this task, we noticed a globally higher sequencing noise, resulting in more reads being removed by quality control. Furthermore, particular care must be given to read trimming due to the observed presence of sequencing artefacts in the reads at the end of the desired amplicon. Given this error rate profile, we suggest that the 97% de-facto standard in OTU picking identity threshold may be too strict and produce some degree of false positive OTU identification that can be mitigated by very low-abundance OTU removal (e.g. 10 sequences for OTU). Nevertheless, we detected a good reproducibility of the results (biological replicates samples 1 and 2 were found to be in strong agreement both by taxonomic composition and by beta-diversity analysis) and comparing our results with other studies (Junemann et al., 2012; Segata et al., 2012), highlighted a good overall consistency and correspondence of the most abundant taxonomic components. Overall, given its ability to produce sequencing results in a short time and with a throughput allowing for a reasonable level of sample multiplexing, IonTorrent appears to be a good compromise for 16S rRNA sequencing and a valid alternative to Illumina 16S rRNA sequencing (Caporaso et al., 2012).

We observed a significant change in microbial community related to short-term probiotic intake, with *Steptococcus* and *Actinomyces* as the most involved genera. The hypothesis we formulated is of an enhanced complexity due to the increased abundance of low abundance organisms occupying the same metabolic niche as the main components of the control saliva microbiome. This hypothesis is further substantiated by the absence of significant changes detected in the metabolic structure of probiotic versus control samples. Interestingly, we did not detect the presence of the administered *Lactobacillus* strains, supporting the hypothesis that colonization is not a prerequisite for microbiome modulation, as non-persistent (passenger) members could induce ecological shifts in the community. However, it should be stressed that this is a short-term probiotic intake (administration on sample collection day); further confirmations should be obtained with a longer-term probiotic intake experiment (possibly including more samples), and with a longitudinal sampling considering multiple time-points before and after the probiotic administration. It would also be of interest to assess the relative contribution of the nutritional components present in the probiotic product with respect to the probiotic strains for the observed microbiome effect, by designing a double-blind placebo controlled experiment. Moreover, other meta'omic approaches (Segata et al., 2013) with species and subspecies phylogenomic resolution could be applied in order to identify which

specific strains are more involved in the increase of diversity following the probiotic intake.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.jbiotec.2014.03.024.

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